

MIDTERM TEST, FALL 2025 Course: CS365 - Intro. to COA Duration: 60 minutes		Student name: Student Id: Order: Class Id:
Assessors' name and signature	Examiners' name and signature	Total score

Test Id: 1 (This test consists of 10 questions)

Note: No electronic devices (laptops, smart devices, etc except calculators) are allowed.

Questions with one correct answer

Question 1. What is the decimal value of this 8-bit two's complement number 1100 0101 after shifting the number right two bits?

- ☐ 15
 ☐ -49
 ☐ 49
 ☐ -15

Question 2. Suppose that register \$t2 contains 0x00000D00 and register \$t1 contains 0x00003C00. Which of the following MIPS instructions places the value 0x00000C00 in register \$t0?

- ☐ and \$t1, \$t2, \$t0
 ☐ add \$t1, \$t2, \$t0
☐ and \$t0, \$t1, \$t2
 ☐ add \$t0, \$t1, \$t2

Question 3. Which of the following MIPS codes loads the 32-bit constant 0x00300100 into register \$s0?

- ☐ lui \$s0, 256
 addi \$s0,\$s0, 48
 ☐ lui \$s0, 46
 addi \$s0,\$s0, 256
☐ lui \$s0, 48
 addi \$s0,\$s0, 256
 ☐ lw \$s0, 256
 add \$s0,\$s0, 48

Question 4. What decimal number is represented by the following IEEE 754 single precision float?

1100 0001 0101 0110 0000 0000 0000 0000

- ☐ 12.375
 ☐ -12.375
 ☐ 13.375
 ☐ -13.375

Question 5. What is the Verilog code that creates a 32-bit value with the pattern 0101...01?

- ☐ {2{16'b01}}
 ☐ {16{2'b01}}
☐ {8'b1,4'b0}
 ☐ 32'b01

Question 6. Suppose that we have two computers A and B implementing the same instruction set architecture. Computer A has a clock cycle time of 1 ns and a CPI of 2.0 for a program P, and computer B has a clock cycle time of 1.5 ns and a CPI of 1.2 for P. How much faster is computer A than computer B for this program?

- ☐ 1.5 times
 ☐ 1.3 times
 ☐ 1.2 times
 ☐ 0.9 times

Constructed-response questions

Question 7. Assume that \$a0 initially contains 5 and \$v0 is used for the output. Add comments to the following MIPS code and describe in one sentence what it computes.

```
begin:  addi $t0, $zero,0      .....
        addi $t1, $zero,1      .....
loop:   slt $t2, $a0, $t1      .....
        bne $t2, $zero,fin      .....
        add $t0, $t0, $t1      .....
        addi $t1, $t1, 2        .....
        j loop                  .....
fin:    add $v0, $t0, $zero      .....

.....
.....
```

Question 8. What is the MIPS assembly code corresponding to the following machine code?

```
100011010010100000000010010110000 .....
00000001001001000001000000000100000 .....
101011010010100000000010010110000 .....
```

Question 9. Compile the following C program into MIPS assembly code.

```
int func (int n) .....
{ .....
    int i; .....
    for (i=0;i<n;i=i+1) .....
        if (i==n) return i; .....
    } .....
.....
.....
.....
```

Question 10. Construct the full truth table described by the following Verilog module.

```
module func {A, B, C, S, D}; .....
    input A, B, C; .....
    output S, D; .....
    assign S=A^B^C; .....
    assign D=(A&B) | (B&C) | (C&A); .....
endmodule .....
.....
.....
.....
```

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Test Id: 2 (This test consists of 10 questions)

Note: No electronic devices (laptops, smart devices, etc except calculators) are allowed.

Questions with one correct answer

Question 1. What is the decimal value of this 8-bit two's complement number 11100101 after shifting the number left two bits?

- ☐ -148
 ☐ 108
 ☐ 148
 ☐ -108

Question 2. Suppose that register \$t2 contains 0x00000D00 and register \$t1 contains 0x00003C00. Which of the following MIPS instructions places the value 0x00004900 in register \$t0?

- ☐ and \$t1, \$t2, \$t0
 ☐ add \$t1, \$t2, \$t0
☐ and \$t0, \$t1, \$t2
 ☐ add \$t0, \$t1, \$t2

Question 3. Which of the following MIPS codes loads the 32-bit constant 0x01000030 into register \$s0?

- ☐ lui \$s0, 256
 addi \$s0,\$s0, 48
 ☐ lui \$s0, 46
 addi \$s0,\$s0, 256
☐ lui \$s0, 48
 addi \$s0,\$s0, 256
 ☐ lw \$s0, 256
 add \$s0,\$s0, 48

Question 4. What decimal number is represented by the following IEEE 754 single precision float?

1011 1111 0100 0000 0000 0000 0000

- ☐ -0.375
 ☐ -1.75
 ☐ 0.375
 ☐ -0.75

Question 5. What is the Verilog code that creates a 32-bit value with the pattern 0101...01?

- ☐ {2{16'b01}}
 ☐ {2{16'h5555}}
☐ {16'b01,16'b01}
 ☐ 32'b01

Question 6. Suppose that we have two computers A and B implementing the same instruction set architecture. Computer A has a clock cycle time of 2 ns and a CPI of 1.5 for a program P, and computer B has a clock cycle time of 1.5 ns and a CPI of 1.2 for P. How much faster is computer A than computer B for this program?

- ☐ 0.9 times
 ☐ 1.3 times
 ☐ 1.2 times
 ☐ 0.6 times

Constructed-response questions

Question 7. Assume that \$a0 initially contains 7 and \$v0 is used for the output. Add comments to the following MIPS code and describe in one sentence what it computes.

```
begin:  addi $t0, $zero,0      .....
        addi $t1, $zero,2      .....
loop:   slt $t2, $a0, $t1      .....
        bne $t2, $zero,fin     .....
        add $t0, $t0, $t1      .....
        addi $t1, $t1, 1       .....
        j loop                 .....
fin:    add $v0, $t0, $zero     .....

.....
.....
```

Question 8. What is the MIPS assembly code corresponding to the following machine code?

```
100011010010100000000010010110000 .....
00000001001001000001000000000100000 .....
101011010010100000000010010110000 .....
```

Question 9. Compile the following C program into MIPS assembly code.

```
int func (int n) .....
{ .....
    int i; .....
    while (i<n) .....
        {i=i+1;} return i; .....
    } .....
.....
.....
.....
```

Question 10. Construct the full truth table described by the following Verilog module.

```
module func {A, B, C, S, D}; .....
    input A, B, C; .....
    output S, D; .....
    assign S=A&B&C; .....
    assign D=(A^B) | (B^C) | (C^A); .....
endmodule .....
.....
.....
.....
```